DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-400-05 — Replace spool SP04 of 4"-P300-P-1019

Area	A400 Pumps	**Task**	SR Spool replacement
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	P-104A — P-104A centrifugal pump — E 116'-6" N 18'-6" EL 100'-9"		
Equipment	P-104A	**Lines**	4"-P300-P-1019, 3/4"-P300-I-7019, 3/4"-P300-SP-7028
Isometrics	ISO-P-1019 (owner's drawings, attached)		
Permits	General work permit, Hot work permit		

Reason

Wall loss below the owner's minimum found by the last survey on the discharge run to V-102.

Scope of work

1. Isolate per the isolation list; drain, vent and purge 4"-P300-P-1019 between the isolations; confirm gas-free for hot work.
2. Remove spool 4"-P300-P-1019-SP04 (12 items, 17'-8-13/16" of pipe) — cut at the field welds and break the flanged joints at its ends as listed.
3. Fabricate the replacement spool to ISO-4"-P300-P-1019 with the same items (parts list), welding to WS-01; 11 shop welds.
4. Install the spool; make 4 field welds; new gaskets and studs at flanged ends, tightened per SW-WO-5.
5. NDE per the NDE table; hydrostatic test at 420 psi for 30 minutes with the relief valves blinded (SW-TEST-1).
6. Remove blinds, restore the line-up, hand back.

Affected items: 4"-P300-P-1019/el21, 4"-P300-P-1019/p22, 4"-P300-P-1019/el23, 4"-P300-P-1019/p24, 4"-P300-P-1019/T:3/4"-P300-I-7019, 4"-P300-P-1019/p26, 4"-P300-P-1019/T:3/4"-P300-SP-7028, 4"-P300-P-1019/p28, 4"-P300-P-1019/el29, 4"-P300-P-1019/p30, 4"-P300-P-1019/el31, 4"-P300-P-1019/p32

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-1, SW-NDE-2, SW-NDE-4, SW-TEST-1, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-4, SW-WO-5, SW-WO-7, SW-WPS-1, SW-WPS-2, SW-WPS-3

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1019	4"-P300-P-1019	4"-P300-P-1019	5	agrees with the model

ISO-P-1019 (PDF): ../isometrics-imported/ISO-P-1019.pdf — the sheets that hold the work follow the ITP.

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
P-104A	equipment	P-104A centrifugal pump		E 116'-6" N 18'-6" EL 100'-9"
4"-P300-P-1019/el21	item	ELBOW 90° R = 5/3 OD 4" P300	4"-P300-P-1019	E 53'-9-5/16" N 5'-0" EL 116'-9-5/16"
4"-P300-P-1019/p22	item	PIPE 4" P300 BEVELLED ENDS	4"-P300-P-1019	E 53'-6" N 5'-0" EL 117'-9"
4"-P300-P-1019/el23	item	ELBOW 90° R = 5/3 OD 4" P300	4"-P300-P-1019	E 53'-6" N 4'-8-11/16" EL 118'-8-11/16"
4"-P300-P-1019/p24	item	PIPE 4" P300 BEVELLED ENDS	4"-P300-P-1019	E 53'-6" N 4'-1-5/16" EL 119'-0"
4"-P300-P-1019/T:3/4"-P300-I-7019	item	TEE REDUCING 4" × 3/4" P300	4"-P300-P-1019	E 53'-6" N 3'-6" EL 119'-0"
4"-P300-P-1019/p26	item	PIPE 4" P300 BEVELLED ENDS	4"-P300-P-1019	E 53'-6" N 2'-9" EL 119'-0"
4"-P300-P-1019/T:3/4"-P300-SP-7028	item	TEE REDUCING 4" × 3/4" P300	4"-P300-P-1019	E 53'-6" N 2'-0" EL 119'-0"
4"-P300-P-1019/p28	item	PIPE 4" P300 BEVELLED ENDS	4"-P300-P-1019	E 53'-6" N -4'-1-5/16" EL 119'-0"
4"-P300-P-1019/el29	item	ELBOW 90° R = 5/3 OD 4" P300	4"-P300-P-1019	E 53'-6" N -10'-2-11/16" EL 119'-3-5/16"
4"-P300-P-1019/p30	item	PIPE 4" P300 BEVELLED ENDS	4"-P300-P-1019	E 53'-6" N -10'-6" EL 120'-0"
4"-P300-P-1019/el31	item	ELBOW 90° R = 5/3 OD 4" P300	4"-P300-P-1019	E 53'-6" N -10'-9-5/16" EL 120'-8-11/16"
4"-P300-P-1019/p32	item	PIPE 4" P300 BEVELLED ENDS	4"-P300-P-1019	E 53'-6" N -12'-1-3/4" EL 121'-0"
4"-P300-P-1019/A5	item	VALVE, GATE, WEDGE, OS&Y 3" P300	4"-P300-P-1019	E 116'-6" N 16'-6" EL 105'-2-3/16"
4"-P300-P-1019/A8	item	FLANGE WELD-NECK 3" P300 RAISED FACE	4"-P300-P-1019	E 116'-6" N 16'-6" EL 105'-11-5/8"
4"-P300-P-1019/B16	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1019	E 53'-6" N -14'-4" EL 121'-0"
4"-P300-P-1019/B19	item	FLANGE WELD-NECK 4" P300 RAISED FACE	4"-P300-P-1019	E 53'-6" N -13'-5-1/2" EL 121'-0"
4"-P300-P-1019/W12	weld	Butt weld 4" P300 (field)	4"-P300-P-1019	E 54'-11/16" N 5'-0" EL 116'-6"
4"-P300-P-1019/W13	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N 5'-0" EL 117'-11/16"
4"-P300-P-1019/W14	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N 5'-0" EL 118'-5-5/16"
4"-P300-P-1019/W15	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N 4'-5-5/16" EL 119'-0"
4"-P300-P-1019/W16	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N 3'-9-5/16" EL 119'-0"

Tag	Kind	Description	Line	Location
4"-P300-P-1019/W17	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N 3'-2-11/16" EL 119'-0"
4"-P300-P-1019/W18	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N 2'-3-5/16" EL 119'-0"
4"-P300-P-1019/W19	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N 20-11/16" EL 119'-0"
4"-P300-P-1019/W20	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N -9'-11-5/16" EL 119'-0"
4"-P300-P-1019/W21	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N -10'-6" EL 119'-6-11/16"
4"-P300-P-1019/W22	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N -10'-6" EL 120'-5-5/16"
4"-P300-P-1019/W23	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N -11'-11/16" EL 121'-0"
4"-P300-P-1019/W24	weld	Butt weld 4" P300 (field)	4"-P300-P-1019	E 53'-6" N -13'-2-13/16" EL 121'-0"
3/4"-P300-I-7019/W1	weld	Butt weld 0.75" P300 (field)	3/4"-P300-I-7019	E 53'-6" N 3'-6" EL 119'-3-5/16"
3/4"-P300-SP-7028/W1	weld	Butt weld 0.75" P300 (field)	3/4"-P300-SP-7028	E 53'-6" N 2'-0" EL 119'-3-5/16"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	4"-P300-P-1019/A5	4"-P300-P-1019	Close and lock VALVE, GATE, WEDGE, OS&Y 3" P300 (4"-P300-P-1019/A5)	E 116'-6" N 16'-6" EL 105'-2-3/16"		
2	blind	4"-P300-P-1019/A8	4"-P300-P-1019	Fit a line blind between 4"-P300-P-1019/A5 and FLANGE WELD-NECK 3" P300 RAISED FACE (4"-P300-P-1019/A8), work side	E 116'-6" N 16'-6" EL 105'-9-1/16"		
3	valve	4"-P300-P-1019/B16	4"-P300-P-1019	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1019/B16)	E 53'-6" N -14'-4" EL 121'-0"		
4	blind	4"-P300-P-1019/B19	4"-P300-P-1019	Fit a line blind between 4"-P300-P-1019/B16 and FLANGE WELD-NECK 4" P300 RAISED FACE (4"-P300-P-1019/B19), work side	E 53'-6" N -13'-8-1/4" EL 121'-0"		

Parts

Description	Qty	Replaces
ELBOW 90° R = 5/3 OD 4" P300	4	
PIPE 4" P300 BEVELLED ENDS	6	
TEE REDUCING 4" × 3/4" P300	2	

New welds (SW-WO-4)

Weld	Replaces	On drawing	Line	Size	Wall	Shop/field	Spec	NDE	Location	Welder	Date
WO-400-05-W01	4"-P300-P-1019/W12	ISO-P-1019 sh 2 W12	4"-P300-P-1019	4" P300	0.1875 in	field	WS-01	VT + RT	E 54'-11/16" N 5'-0" EL 116'-6"		
WO-400-05-W02	4"-P300-P-1019/W13	ISO-P-1019 sh 2 W13	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N 5'-0" EL 117'-11/16"		
WO-400-05-W03	4"-P300-P-1019/W14	ISO-P-1019 sh 3 W14	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N 5'-0" EL 118'-5-5/16"		
WO-400-05-W04	4"-P300-P-1019/W15	ISO-P-1019 sh 3 W15	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N 4'-5-5/16" EL 119'-0"		
WO-400-05-W05	4"-P300-P-1019/W16	ISO-P-1019 sh 3 W16	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N 3'-9-5/16" EL 119'-0"		
WO-400-05-W06	4"-P300-P-1019/W17	ISO-P-1019 sh 3 W17	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N 3'-2-11/16" EL 119'-0"		
WO-400-05-W07	4"-P300-P-1019/W18	ISO-P-1019 sh 4 W18	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N 2'-3-5/16" EL 119'-0"		
WO-400-05-W08	4"-P300-P-1019/W19	ISO-P-1019 sh 4 W19	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N 20-11/16" EL 119'-0"		
WO-400-05-W09	4"-P300-P-1019/W20	ISO-P-1019 sh 4 W20	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N -9'-11-5/16" EL 119'-0"		
WO-400-05-W10	4"-P300-P-1019/W21	ISO-P-1019 sh 4 W21	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N -10'-6" EL 119'-6-11/16"		
WO-400-05-W11	4"-P300-P-1019/W22	ISO-P-1019 sh 5 W22	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N -10'-6" EL 120'-5-5/16"		
WO-400-05-W12	4"-P300-P-1019/W23	ISO-P-1019 sh 5 W23	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N -11'-11/16" EL 121'-0"		
WO-400-05-W13	4"-P300-P-1019/W24	ISO-P-1019 sh 5 W24	4"-P300-P-1019	4" P300	0.1875 in	field	WS-01	VT + RT	E 53'-6" N -13'-2-13/16" EL 121'-0"		

Weld	Replaces	On drawing	Line	Size	Wall	Shop/field	Spec	NDE	Location	Welder	Date
WO-400-05-W14	3/4"-P300-I-7019/W1	ISO-P-1019 sh 3 I-7019/W1	3/4"-P300-I-7019	0.75" P300	0.09375 in	field	WS-01	VT + RT	E 53'-6" N 3'-6" EL 119'-3-5/16"		
WO-400-05-W15	3/4"-P300-SP-7028/W1	ISO-P-1019 sh 4 SP-7028/W1	3/4"-P300-SP-7028	0.75" P300	0.09375 in	field	WS-01	VT + RT	E 53'-6" N 2'-0" EL 119'-3-5/16"		

Welding specification WS-01 — Butt welds, pipe grade, wall to 1/2 in

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Scope	Pipe, fittings and weld-neck flanges of Grade 35 pipe / Grade 36 forging, wall up to 1/2 in
Joint	Single-V, 60° included, root opening 1/8 in, root face 1/16 in (SW-WD-4)
Process	Root: gas-shielded tungsten arc. Fill and cap: shielded metal arc.
Filler	Root SW-F1 wire; fill and cap SW-F2 covered electrode (matching strength, low hydrogen)
Preheat	≥ 30 °F above dew point and ≥ 60 °F, 3 in each side
Interpass	≤ 450 °F
Post-weld heat treatment	Not specified (owner to state)
Rules	SW-WPS-1, SW-WPS-2, SW-WPS-3, SW-WD-4

NDE

Method	Extent	Welds	Basis
VT	100 % of new welds, before and after welding	WO-400-05-W01, WO-400-05-W02, WO-400-05-W03, WO-400-05-W04, WO-400-05-W05, WO-400-05-W06, WO-400-05-W07, WO-400-05-W08, WO-400-05-W09, WO-400-05-W10, WO-400-05-W11, WO-400-05-W12, WO-400-05-W13, WO-400-05-W14, WO-400-05-W15	SW-NDE-1
RT (UT where RT is impractical)	100 % of new welds (class H300, high consequence)	WO-400-05-W01, WO-400-05-W02, WO-400-05-W03, WO-400-05-W04, WO-400-05-W05, WO-400-05-W06, WO-400-05-W07, WO-400-05-W08, WO-400-05-W09, WO-400-05-W10, WO-400-05-W11, WO-400-05-W12, WO-400-05-W13, WO-400-05-W14, WO-400-05-W15	SW-NDE-2

Test — Hydrostatic test

Description	Test boundary between the isolations; relief valves blinded or removed
Pressure	420 psi
Hold	30 minutes, no leak, no pressure drop
Basis	SW-TEST-1

Inspection and test plan (SW-ITP-1..3)

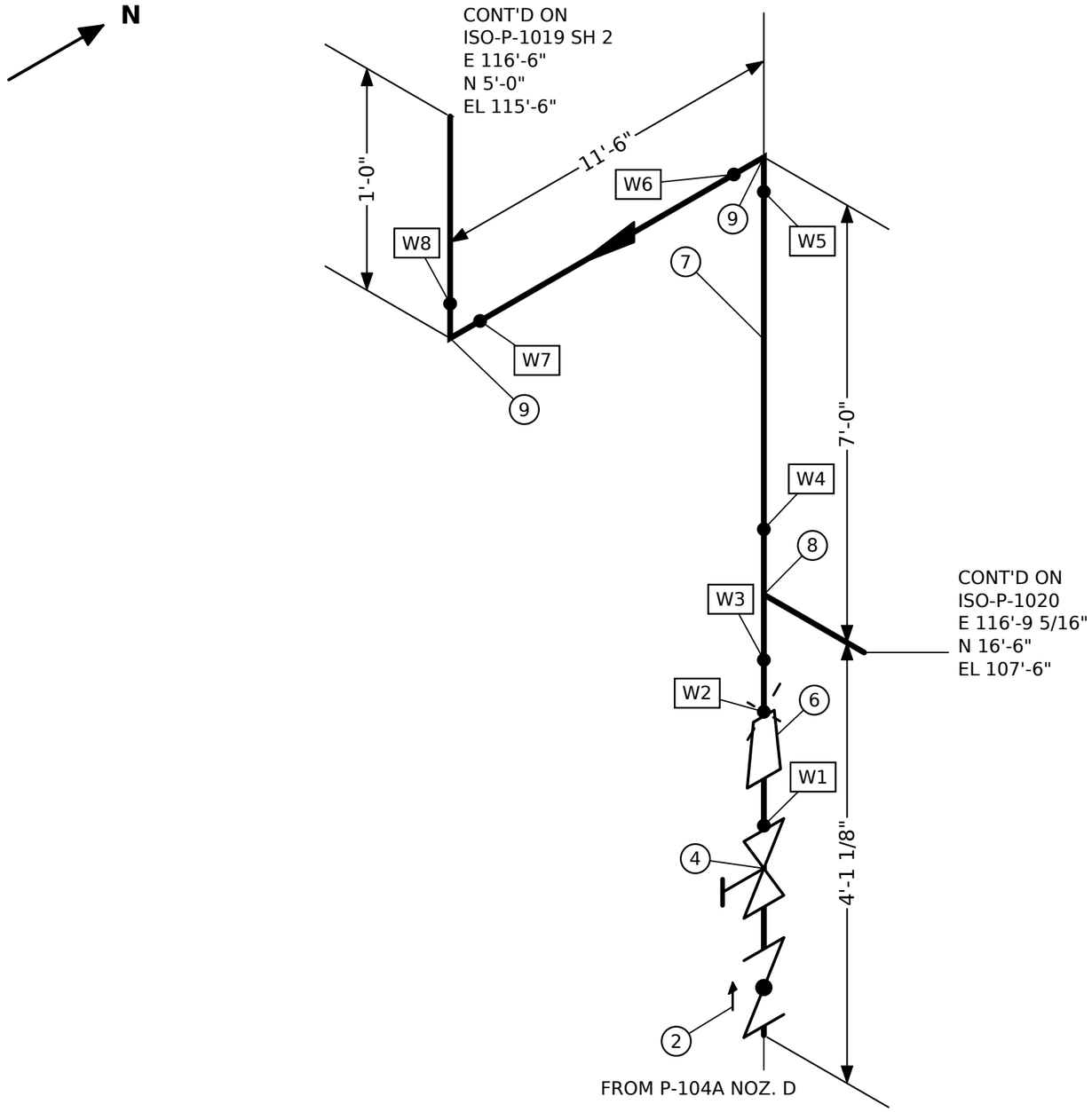
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Line drained, vented and purged; gas-free for hot work	Owner hot-work procedure	Gas test within owner limits	Gas test record, hot work permit	H	H	—				
3	Replacement materials verified against the parts list (size, class, grade, heat marking)	Parts list, SW-LC-3	All items match the line class	Material receiving record	R	R	R				
4	Welding specification WS-01 and welder identification available at the work	WS-01, SW-WPS-3	Specification on site; every welder identified	Welder list	R	R	R				
5	Fit-up of every new weld: root opening and alignment within the root face	SW-WPS-3, SW-WD-4	Root opening 1/8 in; bores aligned within 1/16 in	Fit-up record per weld	H	H	S				
6	Preheat and interpass temperatures maintained during welding	WS-01, SW-WPS-2	Preheat ≥ 30 °F above dew point and ≥ 60 °F, 3 in each side; interpass ≤ 450 °F	Weld record per weld (welder, temperatures)	S	S	S				
7	Visual examination of every new weld	SW-NDE-1, SW-NDE-4	No crack, lack of fusion or incomplete penetration	VT report	W	W	W				
8	Volumetric examination per the NDE table; repairs re-examined	SW-NDE-2, SW-NDE-4	Acceptance per SW-NDE-4 and the owner's criteria	RT/UT reports	R	H	R				
9	Flanged ends: faces inspected, new gaskets, studs to preload	SW-NDE-5, SW-WO-5	No radial scratch; preload as stated	Bolt-up record	W	W	—				
10	Hydrostatic test at 420 psi, 30 minutes	SW-TEST-1	No leak, no pressure drop	Test record with gauge calibration	H	H	H				
11	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
12	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

Tag	Role	Drawing	Sheet	Weld label
4"-P300-P-1019/el21	affected	ISO-P-1019	2	
4"-P300-P-1019/p22	affected	ISO-P-1019	2	
4"-P300-P-1019/el23	affected	ISO-P-1019	3	
4"-P300-P-1019/p24	affected	ISO-P-1019	3	
4"-P300-P-1019/T:3/4"-P300-I-7019	affected	ISO-P-1019	3	
4"-P300-P-1019/p26	affected	ISO-P-1019	3	
4"-P300-P-1019/T:3/4"-P300-SP-7028	affected	ISO-P-1019	4	
4"-P300-P-1019/p28	affected	ISO-P-1019	4	
4"-P300-P-1019/el29	affected	ISO-P-1019	4	
4"-P300-P-1019/p30	affected	ISO-P-1019	4	
4"-P300-P-1019/el31	affected	ISO-P-1019	5	
4"-P300-P-1019/p32	affected	ISO-P-1019	5	
4"-P300-P-1019/W12	replaced by WO-400-05-W01	ISO-P-1019	2	W12
4"-P300-P-1019/W13	replaced by WO-400-05-W02	ISO-P-1019	2	W13
4"-P300-P-1019/W14	replaced by WO-400-05-W03	ISO-P-1019	3	W14
4"-P300-P-1019/W15	replaced by WO-400-05-W04	ISO-P-1019	3	W15
4"-P300-P-1019/W16	replaced by WO-400-05-W05	ISO-P-1019	3	W16
4"-P300-P-1019/W17	replaced by WO-400-05-W06	ISO-P-1019	3	W17
4"-P300-P-1019/W18	replaced by WO-400-05-W07	ISO-P-1019	4	W18
4"-P300-P-1019/W19	replaced by WO-400-05-W08	ISO-P-1019	4	W19
4"-P300-P-1019/W20	replaced by WO-400-05-W09	ISO-P-1019	4	W20
4"-P300-P-1019/W21	replaced by WO-400-05-W10	ISO-P-1019	4	W21
4"-P300-P-1019/W22	replaced by WO-400-05-W11	ISO-P-1019	5	W22
4"-P300-P-1019/W23	replaced by WO-400-05-W12	ISO-P-1019	5	W23
4"-P300-P-1019/W24	replaced by WO-400-05-W13	ISO-P-1019	5	W24
3/4"-P300-I-7019/W1	replaced by WO-400-05-W14	ISO-P-1019	3	I-7019/W1
3/4"-P300-SP-7028/W1	replaced by WO-400-05-W15	ISO-P-1019	4	SP-7028/W1
4"-P300-P-1019/A5	isolation: valve	ISO-P-1019	1	
4"-P300-P-1019/A8	isolation: blind	ISO-P-1019	1	
4"-P300-P-1019/B16	isolation: valve	ISO-P-1019	5	
4"-P300-P-1019/B19	isolation: blind	ISO-P-1019	5	



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	3	3"	GASKET 3" P300 COMPRESSED SHEET
2	1	3"	VALVE, SWING CHECK 3" P300
3	3	3"	STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300
4	1	3"	VALVE, GATE, WEDGE, OS&Y 3" P300
5	1	3"	FLANGE WELD-NECK 3" P300 RAISED FACE
6	1	4"x3"	REDUCER CONCENTRIC 4" × 3" P300
7	17'-8 1/8"	4"	PIPE 4" P300 BEVELLED ENDS
8	1	4"	TEE EQUAL 4" P300
9	2	4"	ELBOW 90° R = 5/3 OD 4" P300

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

NOTES:

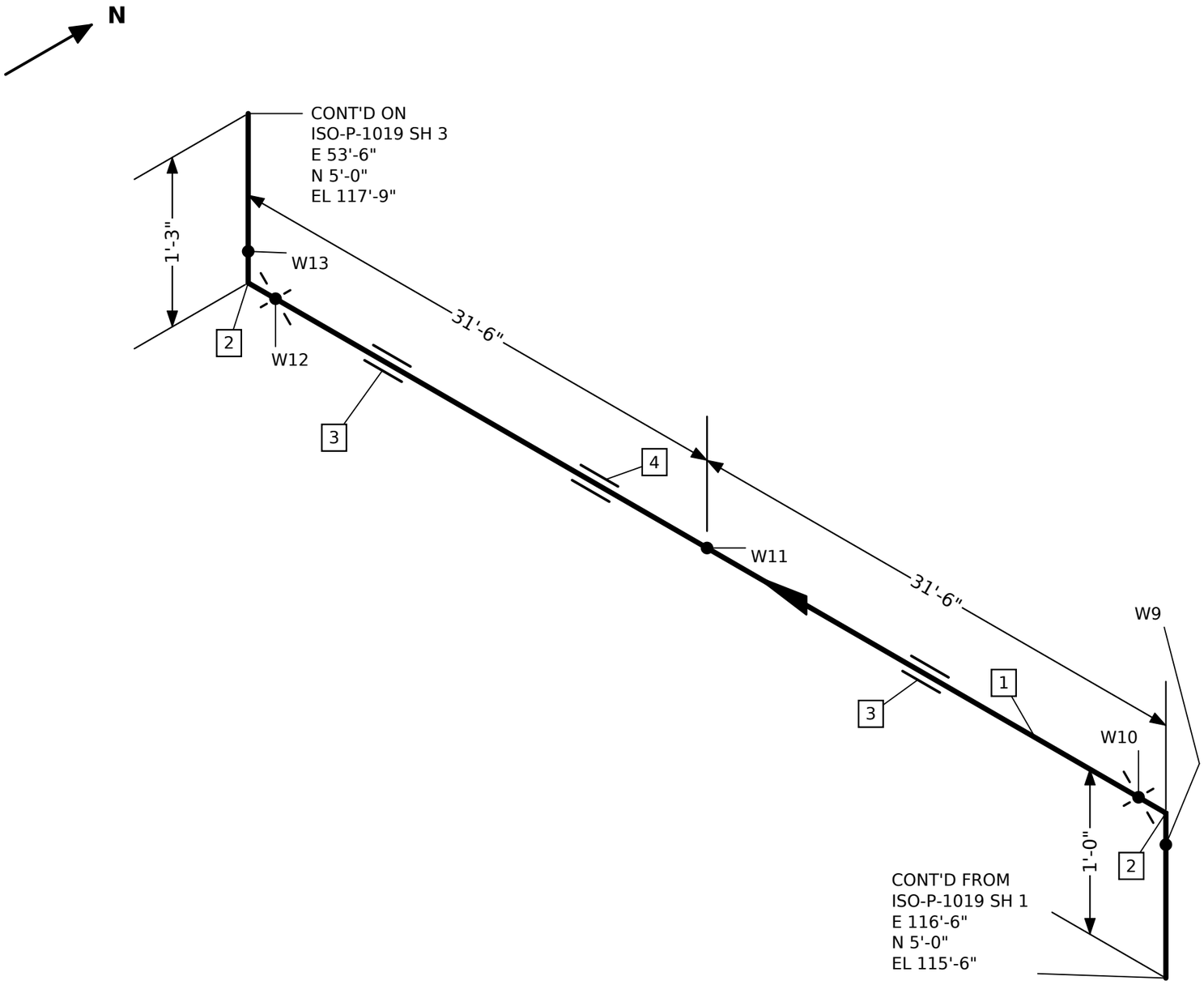
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. PAINT PER PROJECT SPECIFICATION

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	P

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P300-P-1019	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1019	1 OF 5	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	63'-5/16"	4"	PIPE 4" P300 BEVELLED ENDS
2	2	4"	ELBOW 90° R = 5/3 OD 4" P300
3	2	4"	GUIDE FOR 4" LINE (GUIDE-4)
4	1	4"	ANCHOR FOR 4" LINE (ANCHOR-4)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

NOTES:
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. REMOVE BURRS FROM ALL CUT ENDS
4. CHECK ORIENTATION BEFORE WELDING

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	P

COMPANY

SMARTWELD

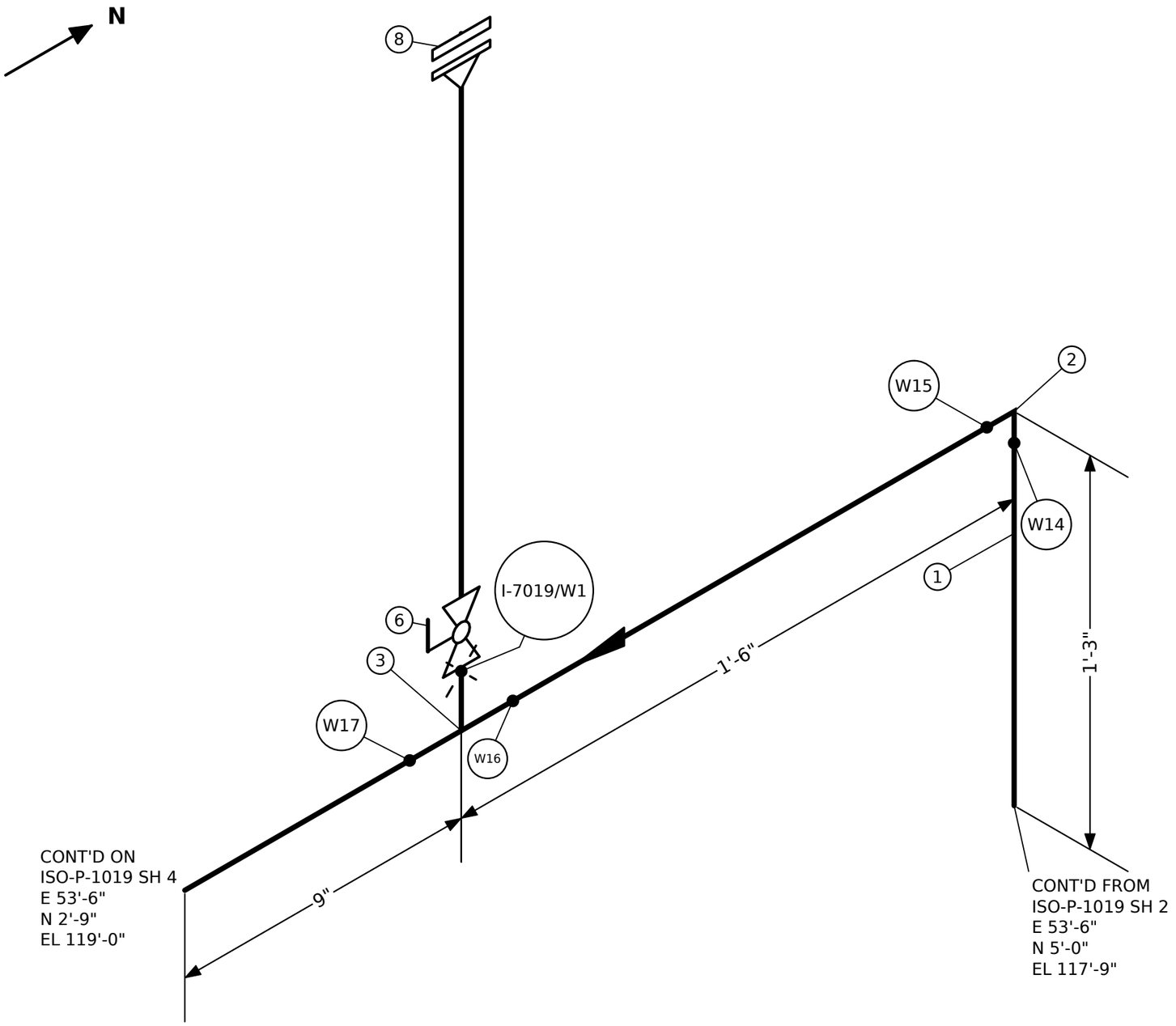
TITLE

HYDROCARBON PROCESS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P300-P-1019	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1019	2 OF 5	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	1'-10"	4"	PIPE 4" P300 BEVELLED ENDS
2	1	4"	ELBOW 90° R = 5/3 OD 4" P300
3	1	4"x3/4"	TEE REDUCING 4" × 3/4" P300
4	1	3/4"	FLANGE WELD-NECK 3/4" P300 RAISED FACE
5	2	3/4"	GASKET 3/4" P300 COMPRESSED SHEET
6	1	3/4"	VALVE, BALL, FLOATING 3/4" P300
7	2	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300
8	1	3/4"	FLANGE BLIND 3/4" P300 RAISED FACE

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

TEST PRESS. -	INSULATION NONE	PAINT -	
PWHT -	NDE -	FLUID P	

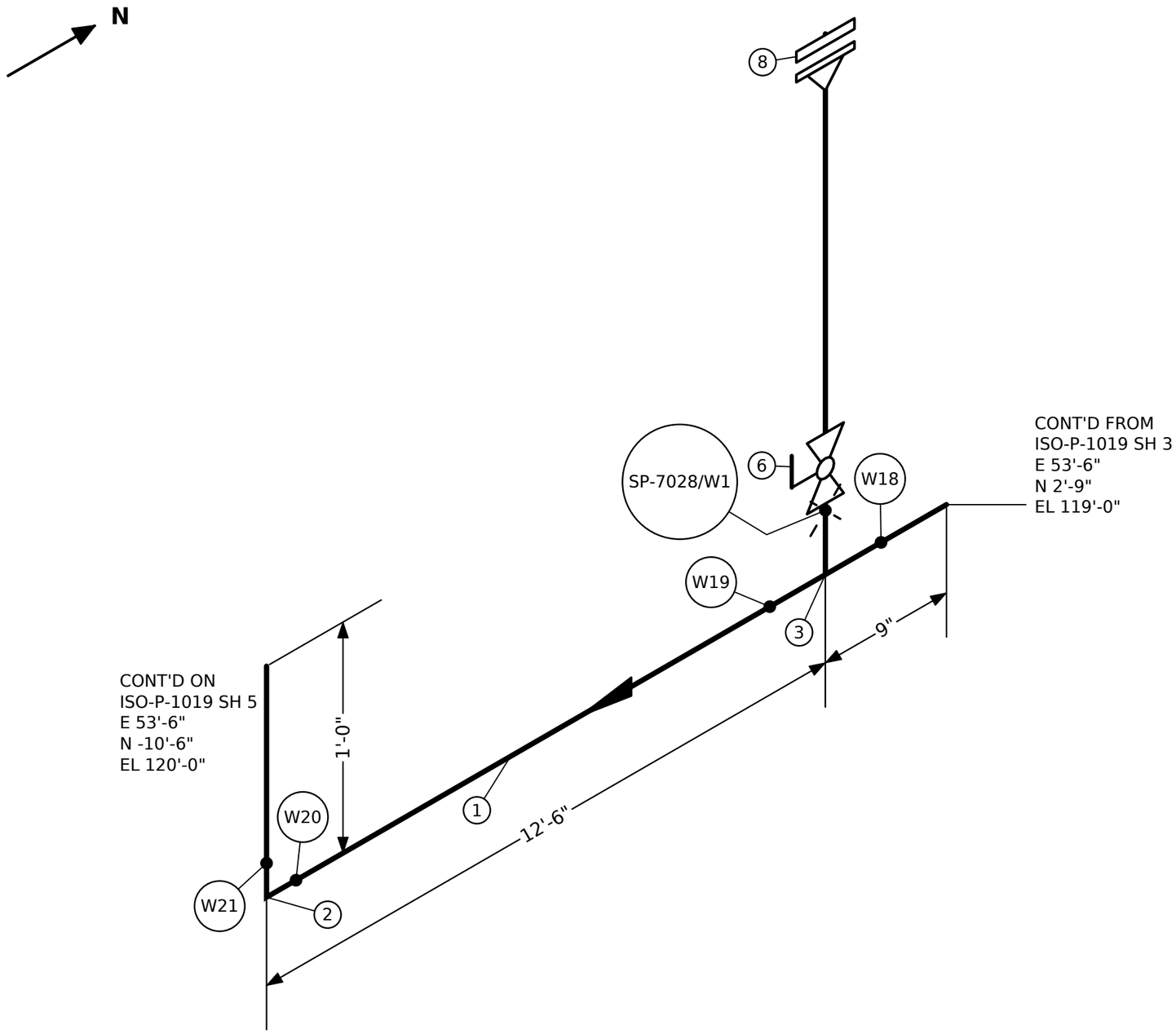
COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

LINE NO. 4"-P300-P-1019	PIPING CLASS H300	DESIGN PRESS. 300 PSIG	DESIGN TEMP. -
DRAWING NO. ISO-P-1019		SHEET 3 OF 5	REV 1
DATE 2026-11-01	DRAWN -	CHECKED -	SCALE NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	12'-7"	4"	PIPE 4" P300 BEVELLED ENDS
2	1	4"	ELBOW 90° R = 5/3 OD 4" P300
3	1	4"x3/4"	TEE REDUCING 4" × 3/4" P300
4	1	3/4"	FLANGE WELD-NECK 3/4" P300 RAISED FACE
5	2	3/4"	GASKET 3/4" P300 COMPRESSED SHEET
6	1	3/4"	VALVE, BALL, FLOATING 3/4" P300
7	2	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300
8	1	3/4"	FLANGE BLIND 3/4" P300 RAISED FACE

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. FLANGE BOLT HOLES STRADDLE CENTRELINES
4. CHECK ORIENTATION BEFORE WELDING

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	P

COMPANY

SMARTWELD

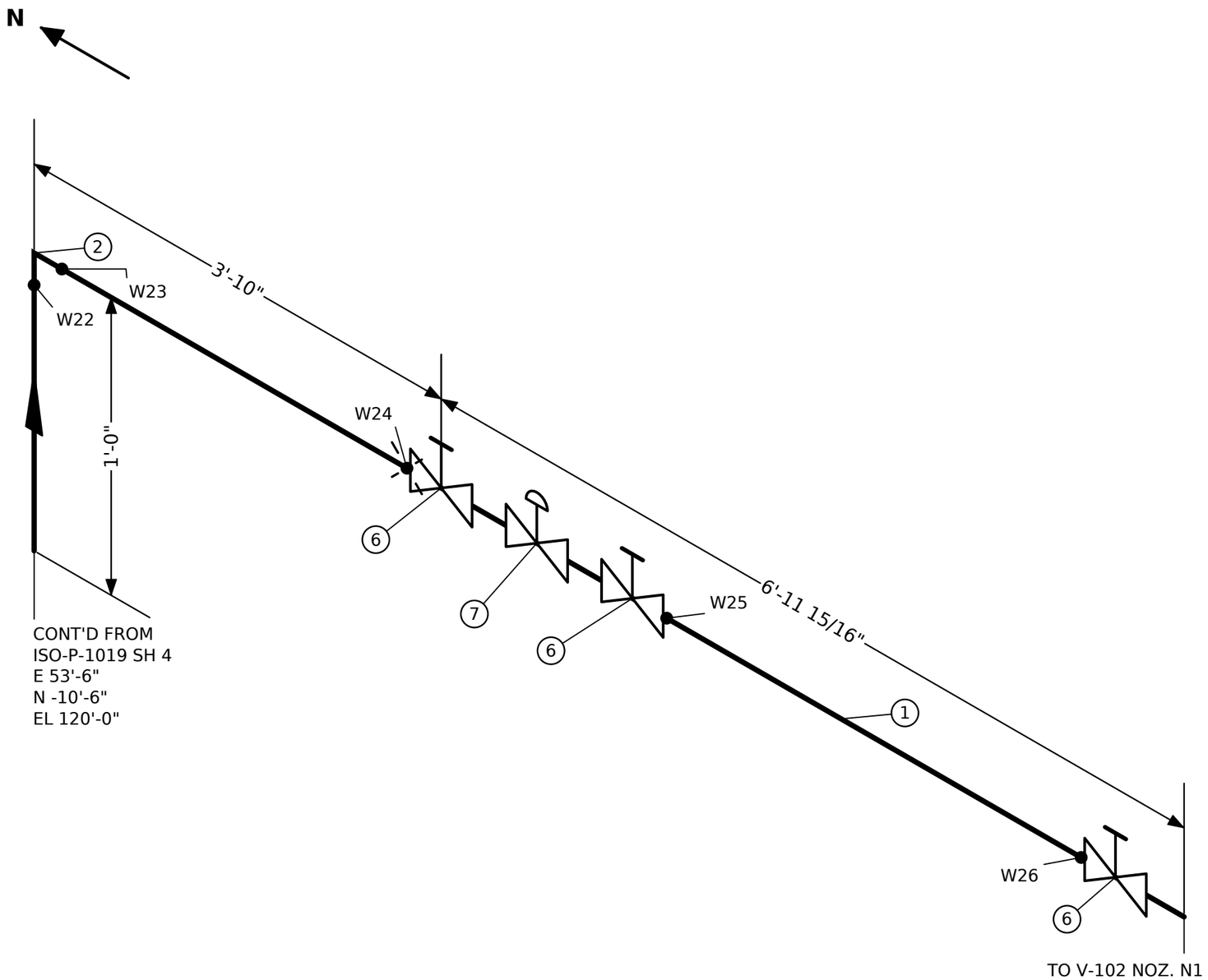
TITLE

HYDROCARBON PROCESS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P300-P-1019	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1019	4 OF 5	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	4'-1 1/2"	4"	PIPE 4" P300 BEVELLED ENDS
2	1	4"	ELBOW 90° R = 5/3 OD 4" P300
3	6	4"	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300
4	3	4"	FLANGE WELD-NECK 4" P300 RAISED FACE
5	6	4"	GASKET 4" P300 COMPRESSED SHEET
6	3	4"	VALVE, GATE, WEDGE, OS&Y 4" P300
7	1	4"	VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. FLANGE BOLT HOLES STRADDLE CENTRELINES
4. PAINT PER PROJECT SPECIFICATION

TEST PRESS.	INSULATION	PAINT
-	NONE	-

PWHT	NDE	FLUID
-	-	P

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P300-P-1019	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1019	5 OF 5	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS

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